Tema 11: Potencial de acción de las fibras nerviosas.

Facultad de Biología. Grado de Bioquímica.

Asignatura de “Biofísica”. Dpto. de Fisiología Médica y Biofísica.

Prof.: Rafael J. Montoro

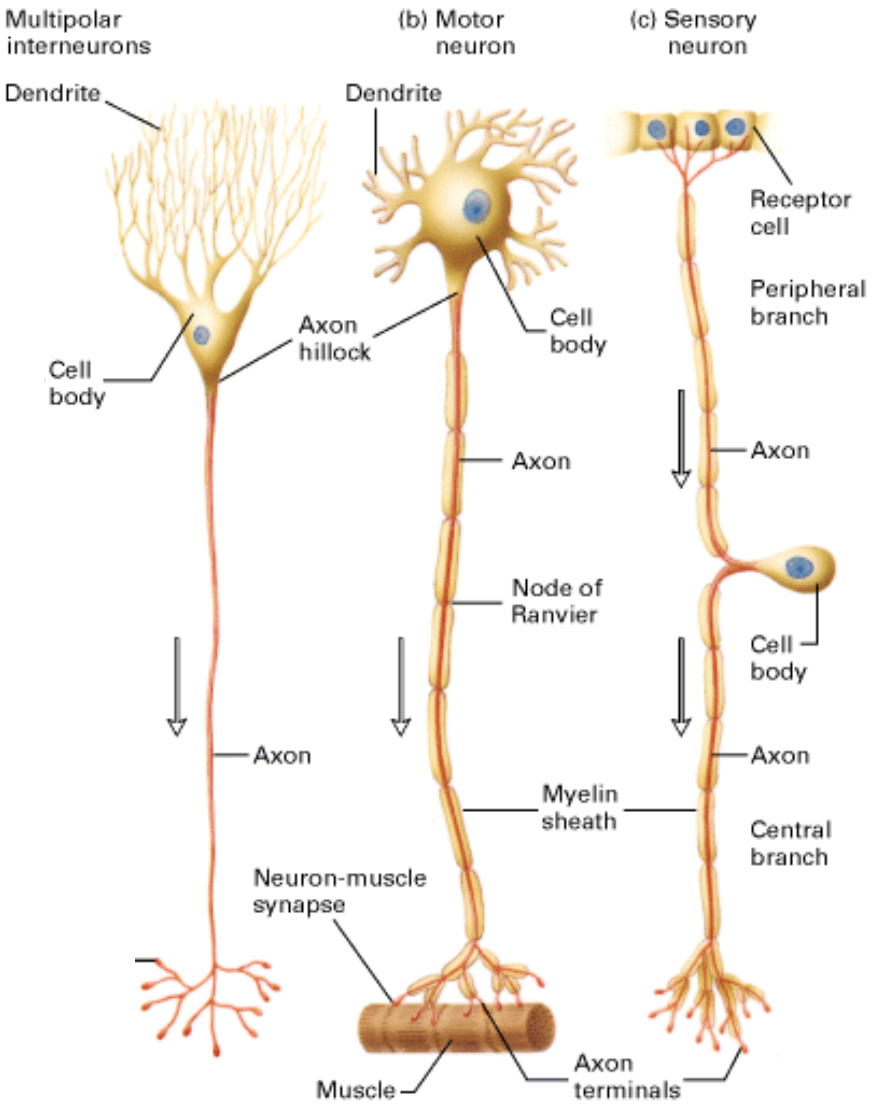
Tema 11. Potencial de acción de las fibras nerviosas. Bases iónicas y moleculares del impulso nervioso.

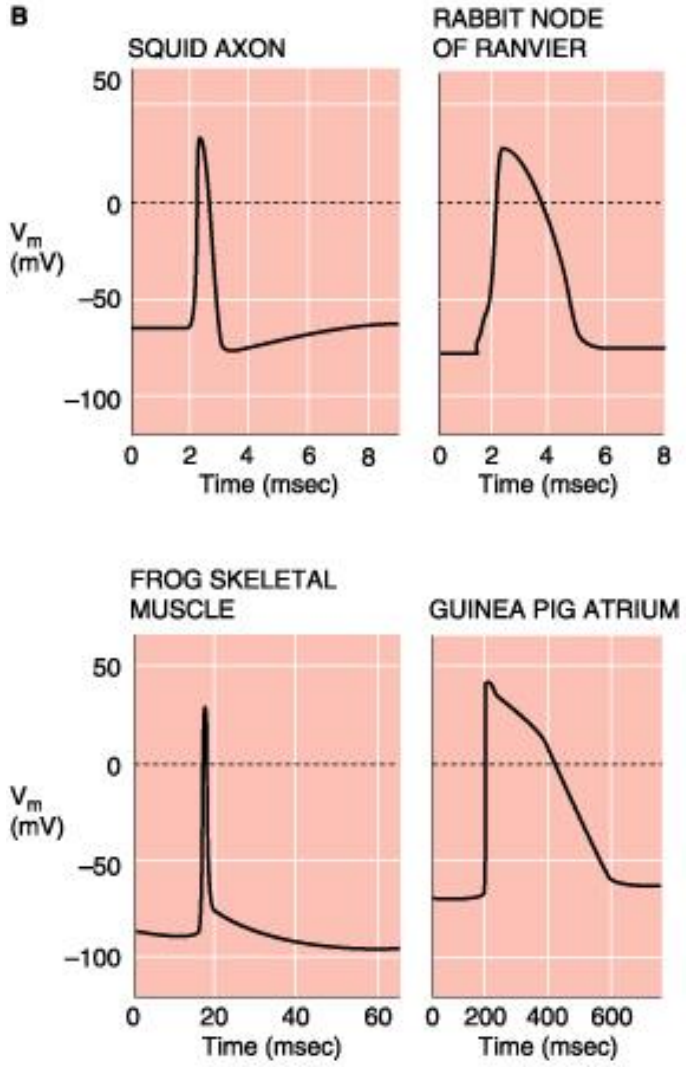
- Potencial de acción
  - concepto
  - fases
    - despolarización umbral
    - fase de despolarización
    - potencial de inversión
    - repolarización y postpotenciales
  - bases iónicas y moleculares
    - experimentos de sustitución iónica

- Registro de las corrientes iónicas transmembrana
  - técnica de fijación de voltaje o “voltage clamp”
  - técnica de “patch clamp” o fijación de sello
  - corrientes obtenidas
    - corrientes capacitativa y de fuga
    - corrientes voltaje dependientes
      - separación de corrientes
      - curvas I/V
    - conductancias voltaje dependientes
    - corrientes de compuerta

- Características generales de los potenciales de acción
  - umbral de activación
  - respuesta todo o nada
  - período refractario
    - absoluto
    - relativo
  - propagación sin decremento

EL POTENCIAL DE ACCIÓN COMO UNIDAD DE INFORMACIÓN





REGISTRO DEL POTENCIAL DE ACCIÓN

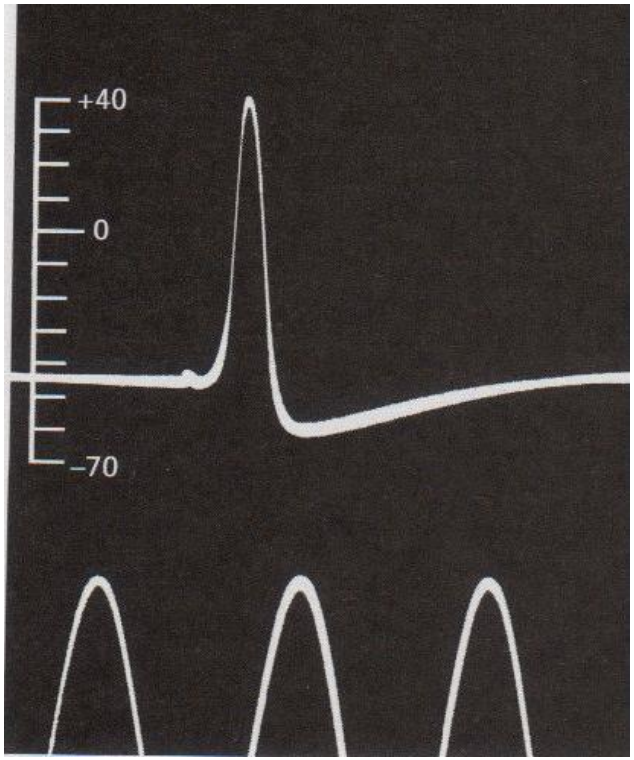
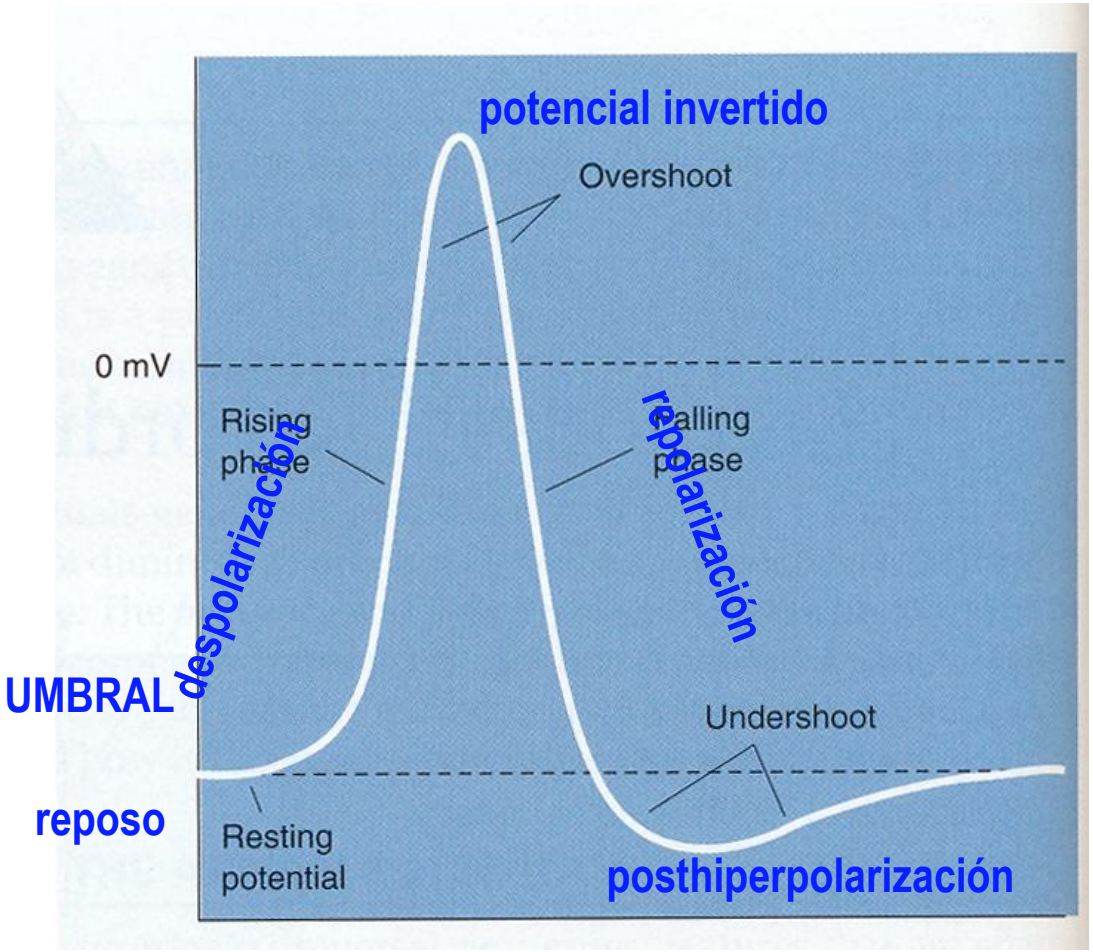


Figure 2-3 This historic tracing is the first published intracellular recording of an action potential. It was obtained in 1939 by Hodgkin and Huxley from the squid giant axon, using glass capillary electrodes filled with sea water. Time marker is 500 Hz. The vertical scale indicates the potential of the internal electrode in millivolts, the sea water outside being taken as zero potential. (From Hodgkin and Huxley 1939.)

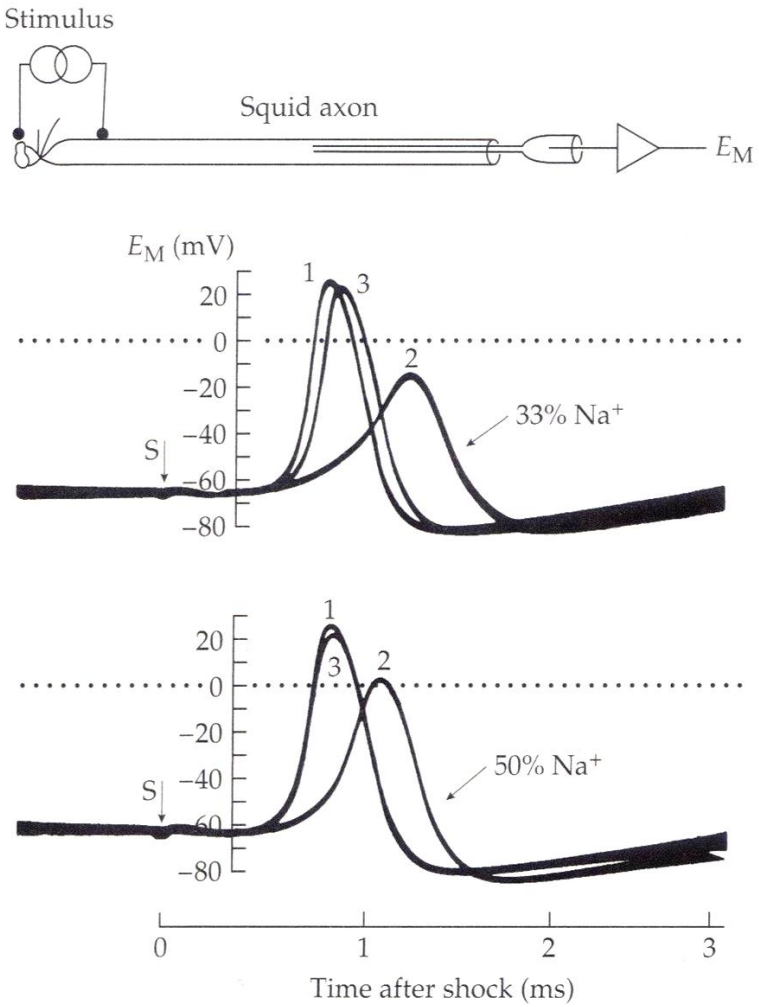
Tomada de E. Kandel. Principles of Neural Science. Fig. 2-3. 4th ed.

Fases del potencial de acción



Tomado de Neuroscience. Exploring the brain
Bears, Connors y Paradiso. 2nd ed

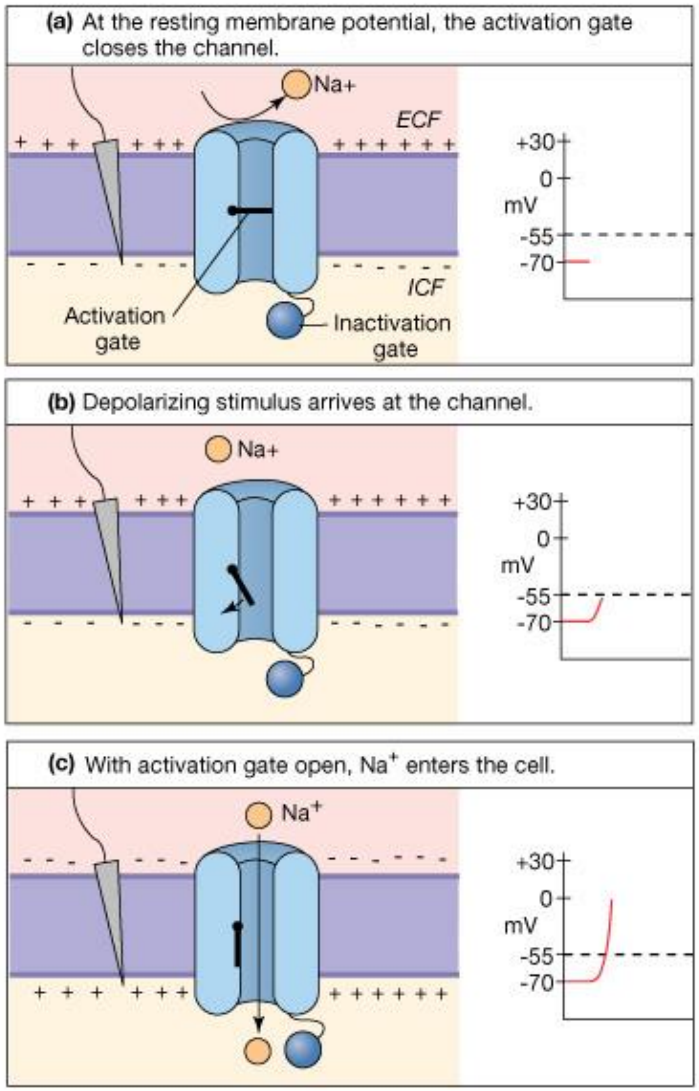
DEPENDENCIA DEL POTENCIAL DE ACCIÓN DEL SODIO EXTRACELULAR

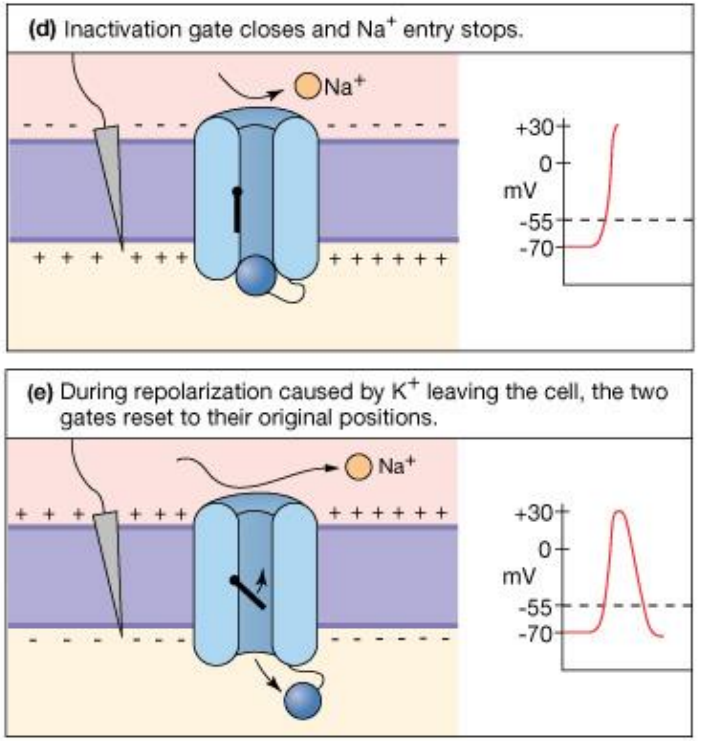


2.4 Na^+-Dependence of the Action Potential This is the first experiment to demonstrate that external Na^+ ions are needed for propagated action potentials. Intracellular potential is recorded with an axial microelectrode inside a squid giant axon. The action potential is smaller and rises more slowly in solutions containing less than the normal amount of Na^+. External bathing solutions: Records 1 and 3 in normal seawater; record 2 in low-sodium solution containing 1:2 or 1:1 mixtures of seawater with isotonic glucose. An assumed 15-mV junction potential has been subtracted from the voltage scale. [From Hodgkin and Katz 1949.]

Tomada de “Ion channels of excitable membranes”. B. Hilll. 3ª ed.

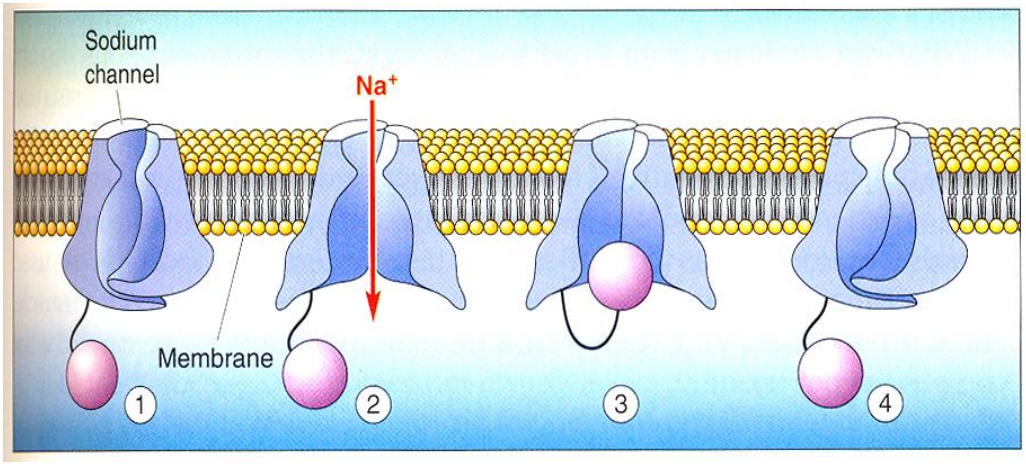
COMPUERTAS DE ACTIVACIÓN E INACTIVACIÓN



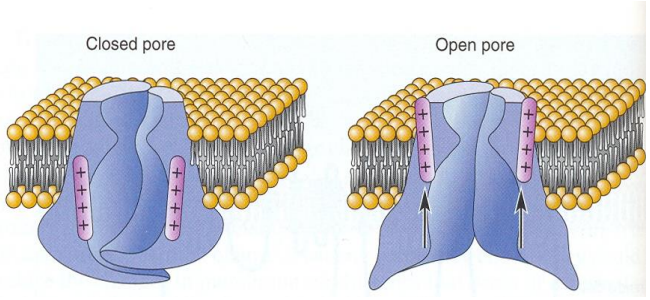


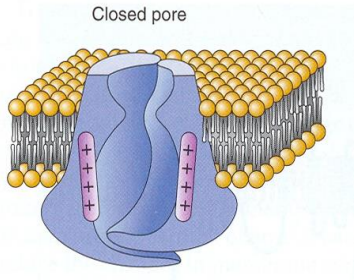
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BASES IÓNICAS Y MOLECULARES DEL POTENCIAL DE ACCIÓN

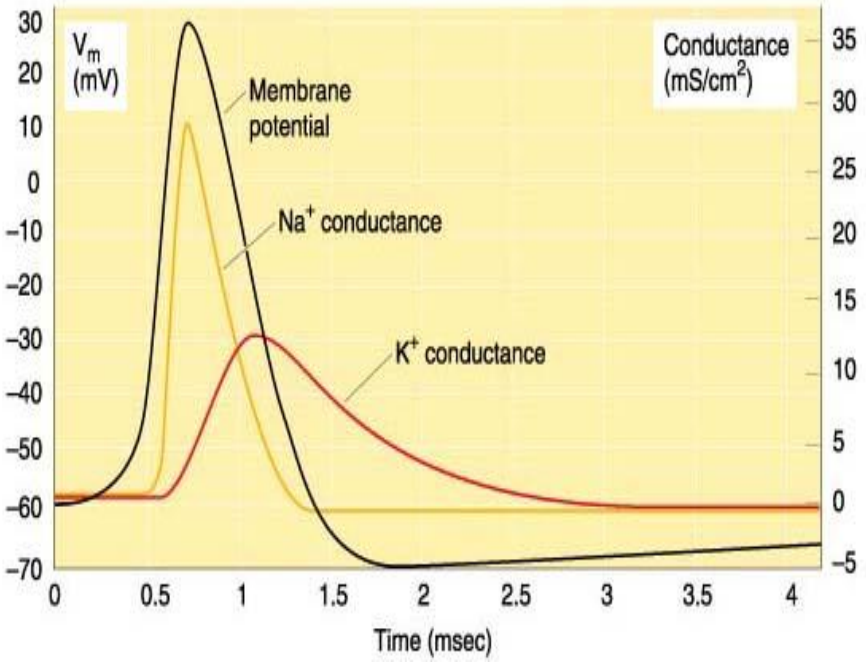


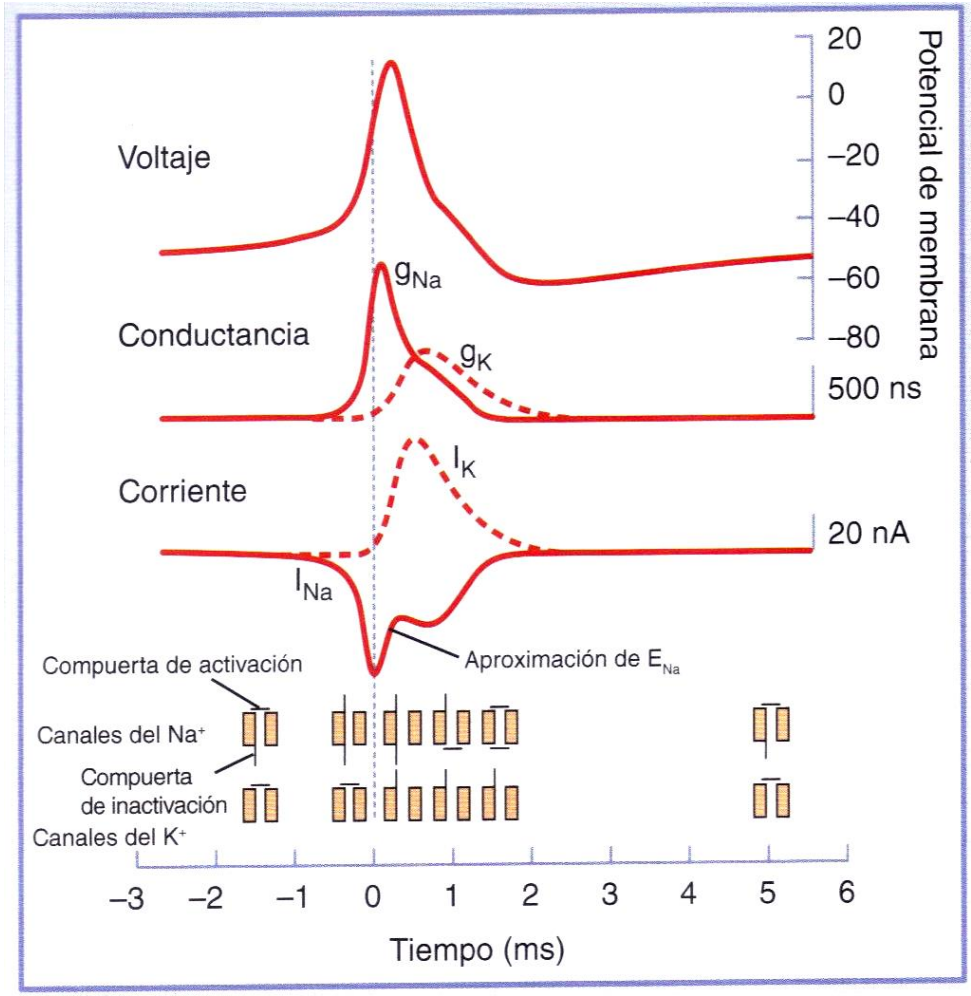
Potassium channel



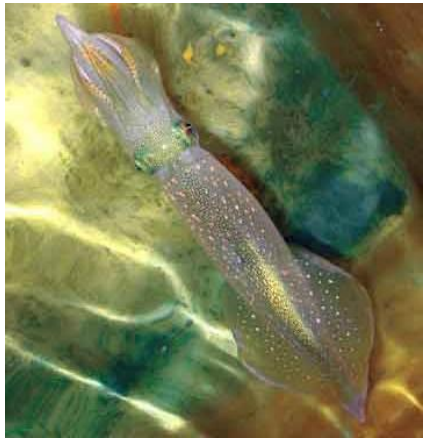


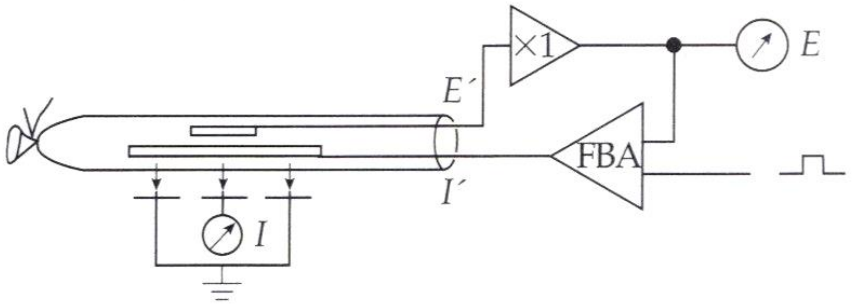
BASES IÓNICAS Y MOLECULARES DEL POTENCIAL DE ACCIÓN

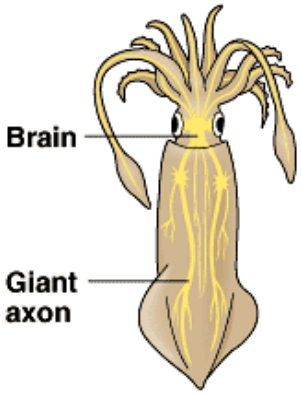




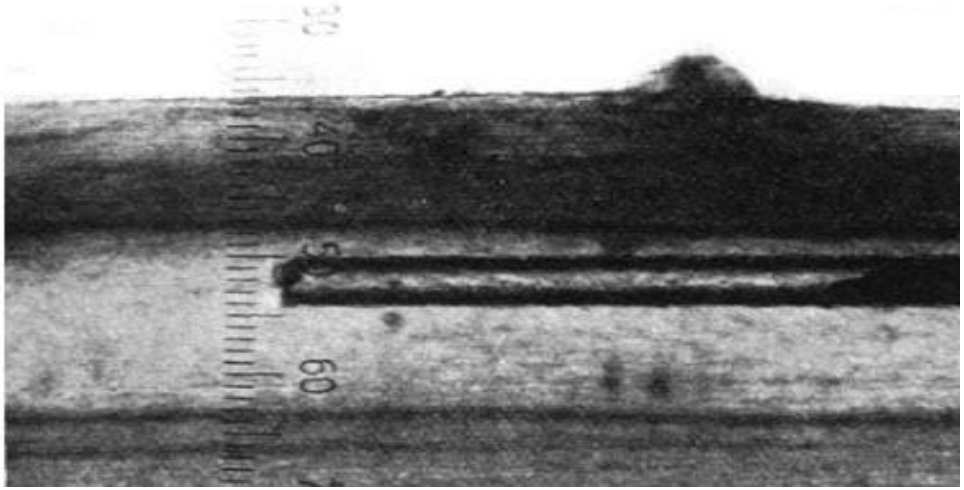
TÉCNICA DE “VOLTAGE CLAMP” (HODGKIN Y HUXLEY)



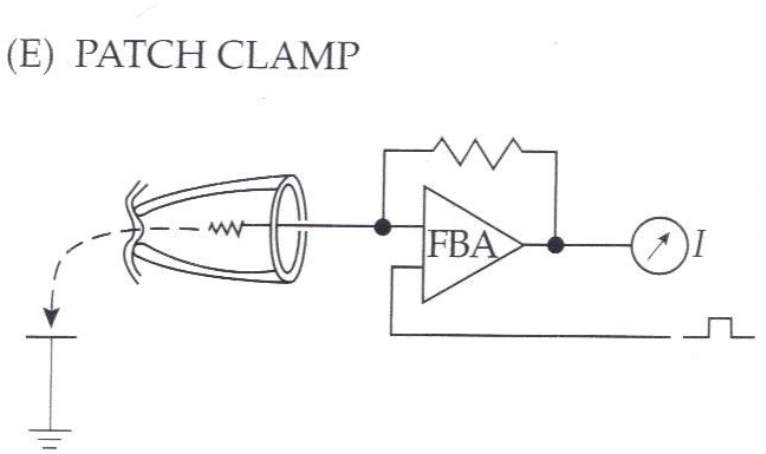


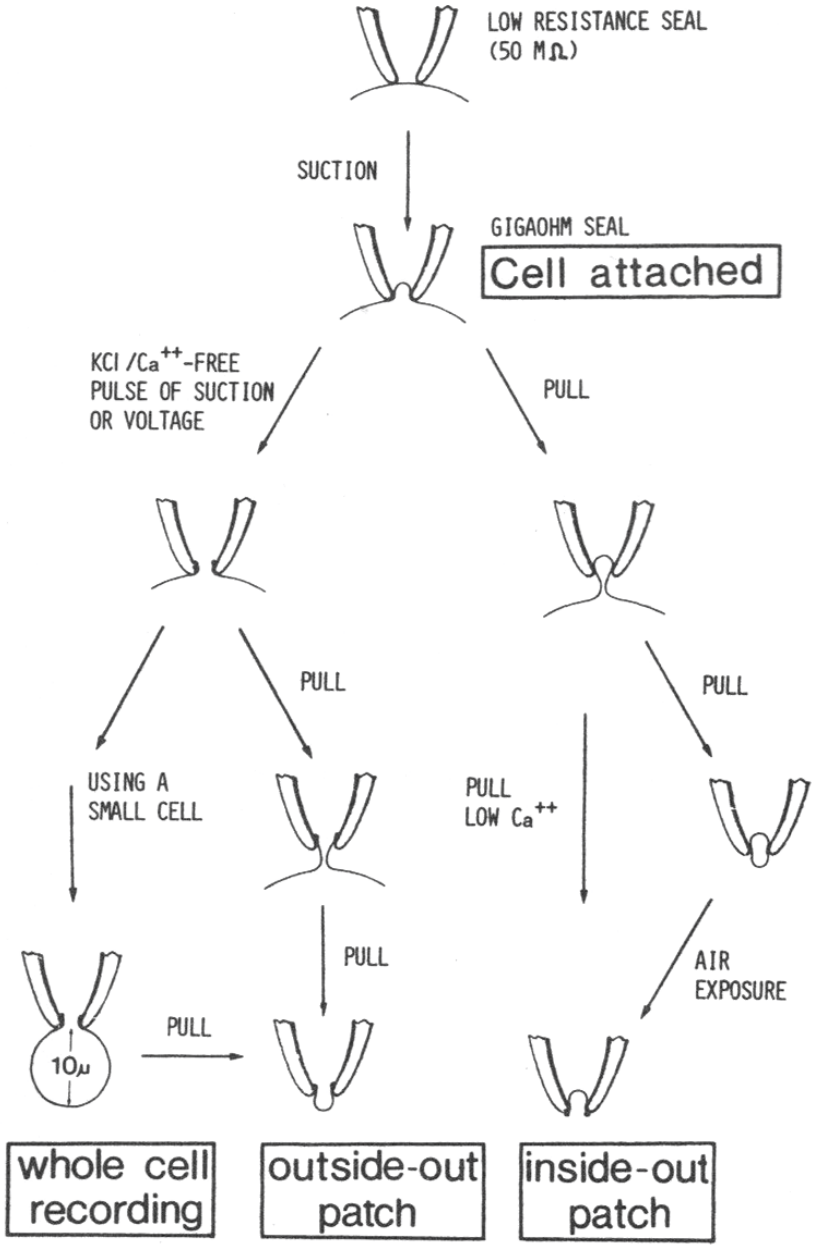


Squid (mollusk)



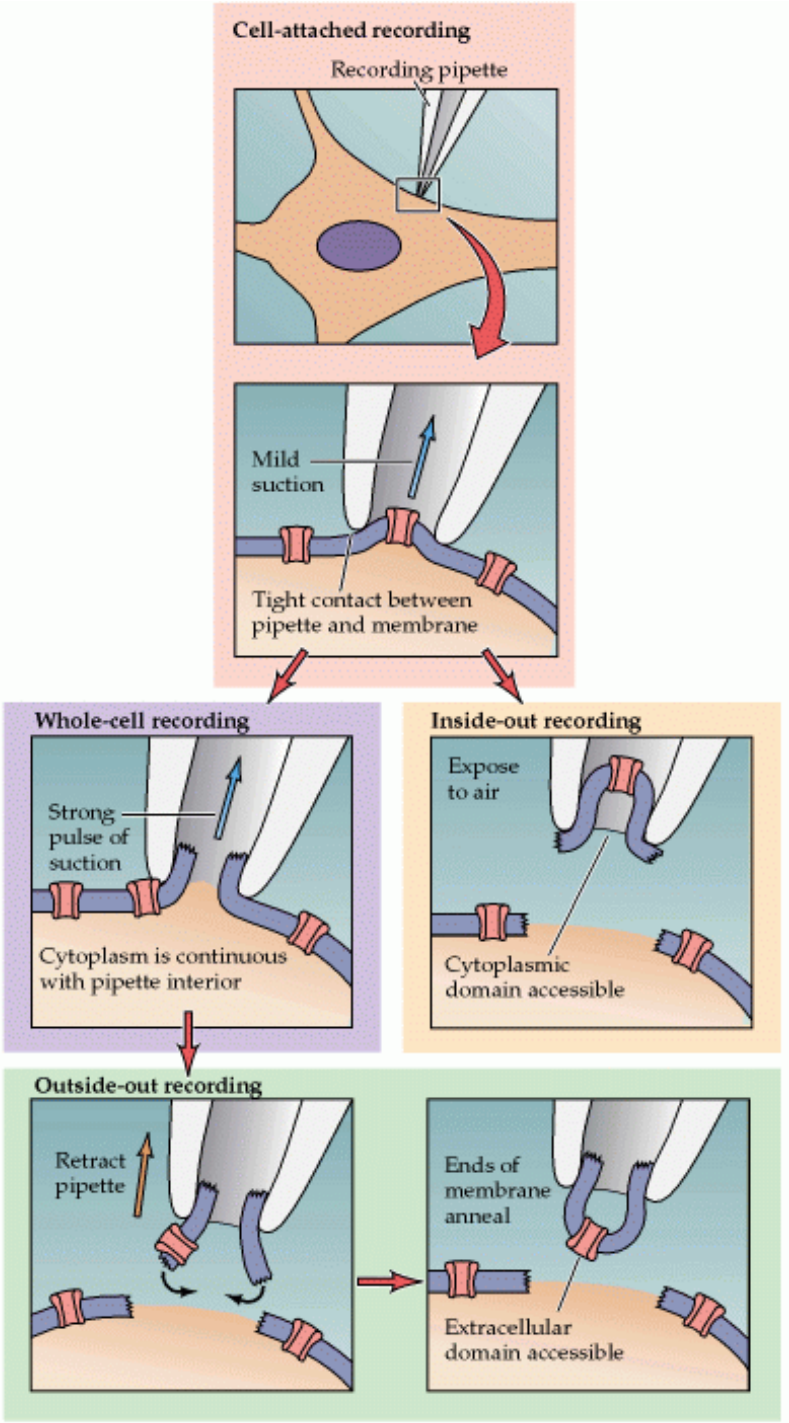
TÉCNICA DE "PATCH CLAMP" (NEHER Y SAKMANN)





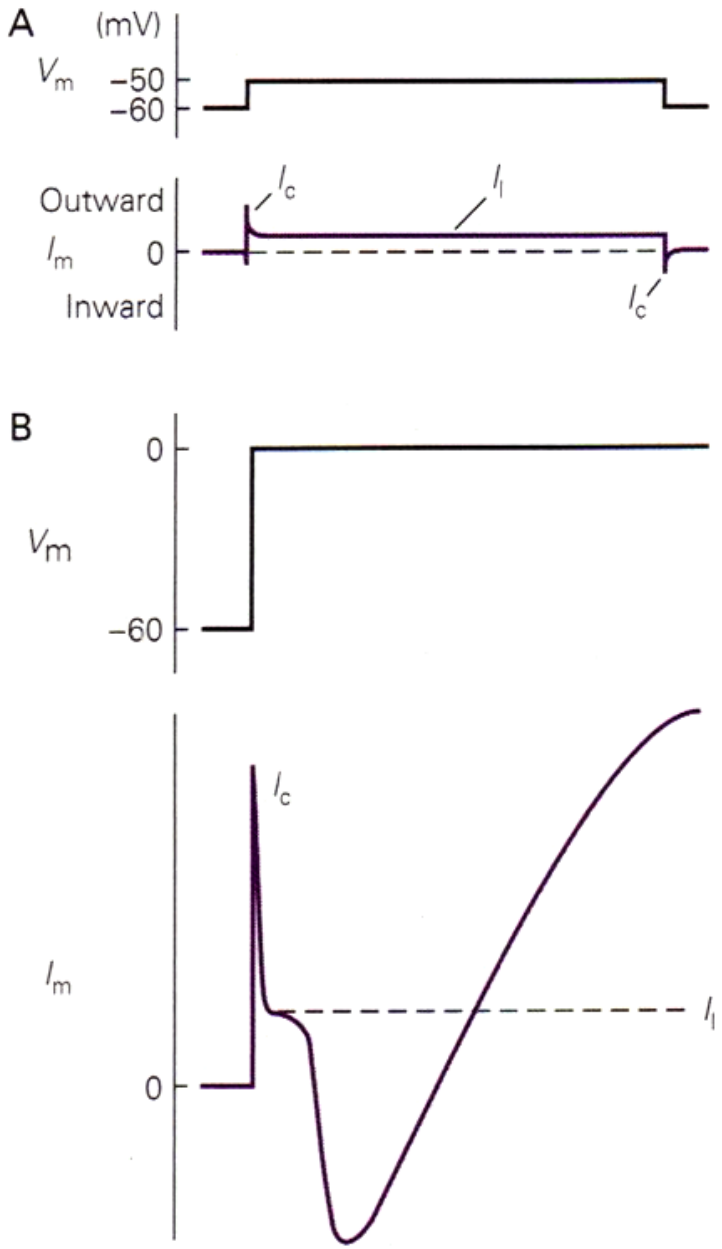
TÉCNICA DE “PATCH CLAMP”
(NEHER Y SAKMANN)

Célula completa

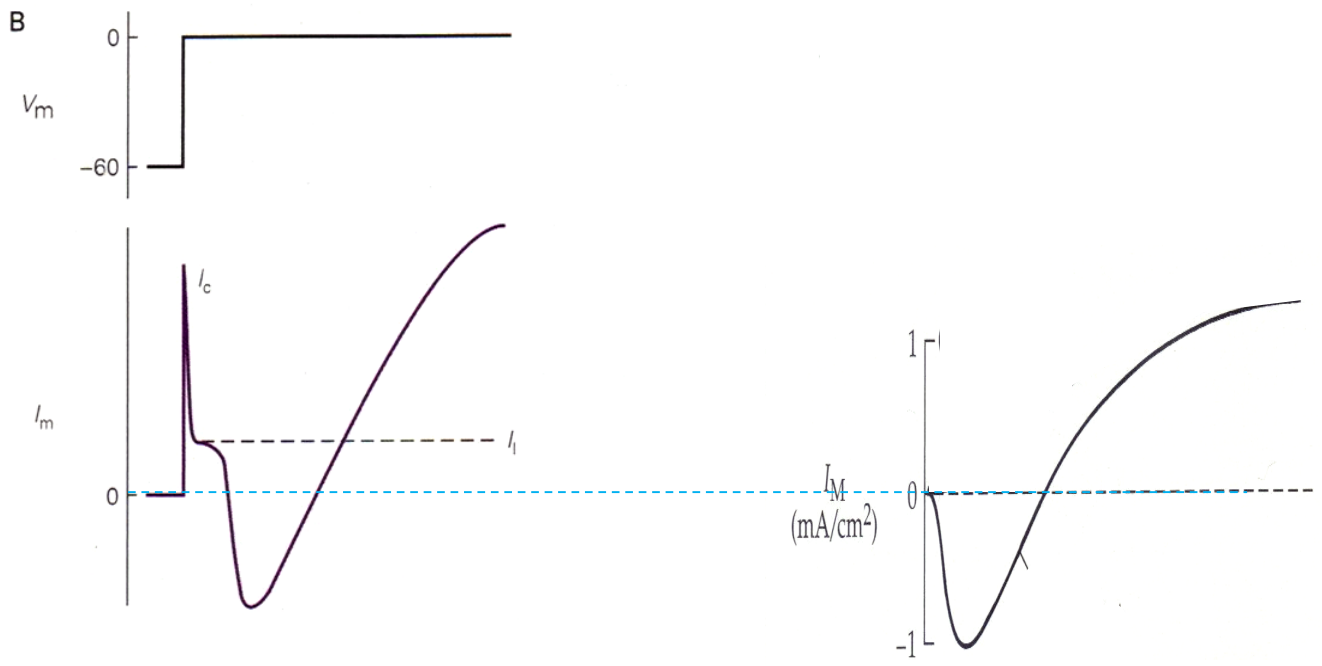


Canal único

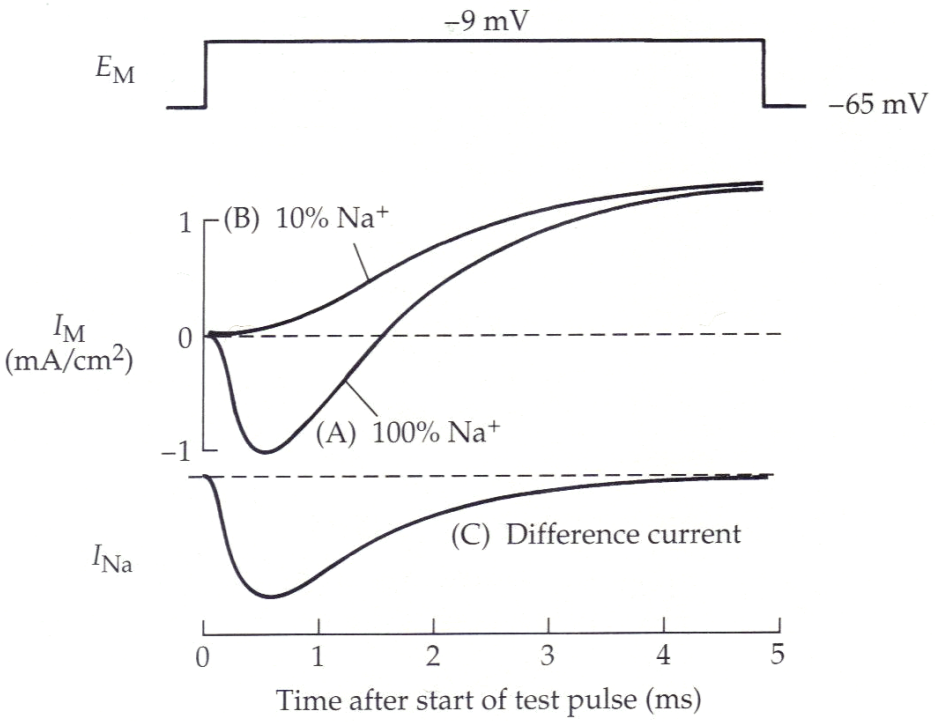
CORRIENTE CAPACITATIVA
Y DE FUGA

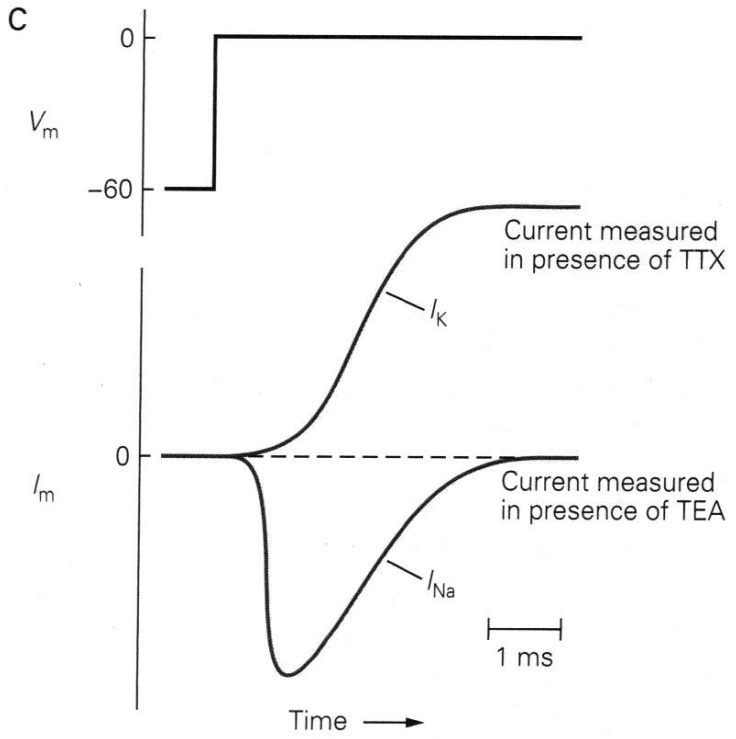


ELIMINACIÓN DE LOS COMPONENETES PASIVOS DEL REGISTRO DE CORRIENTE

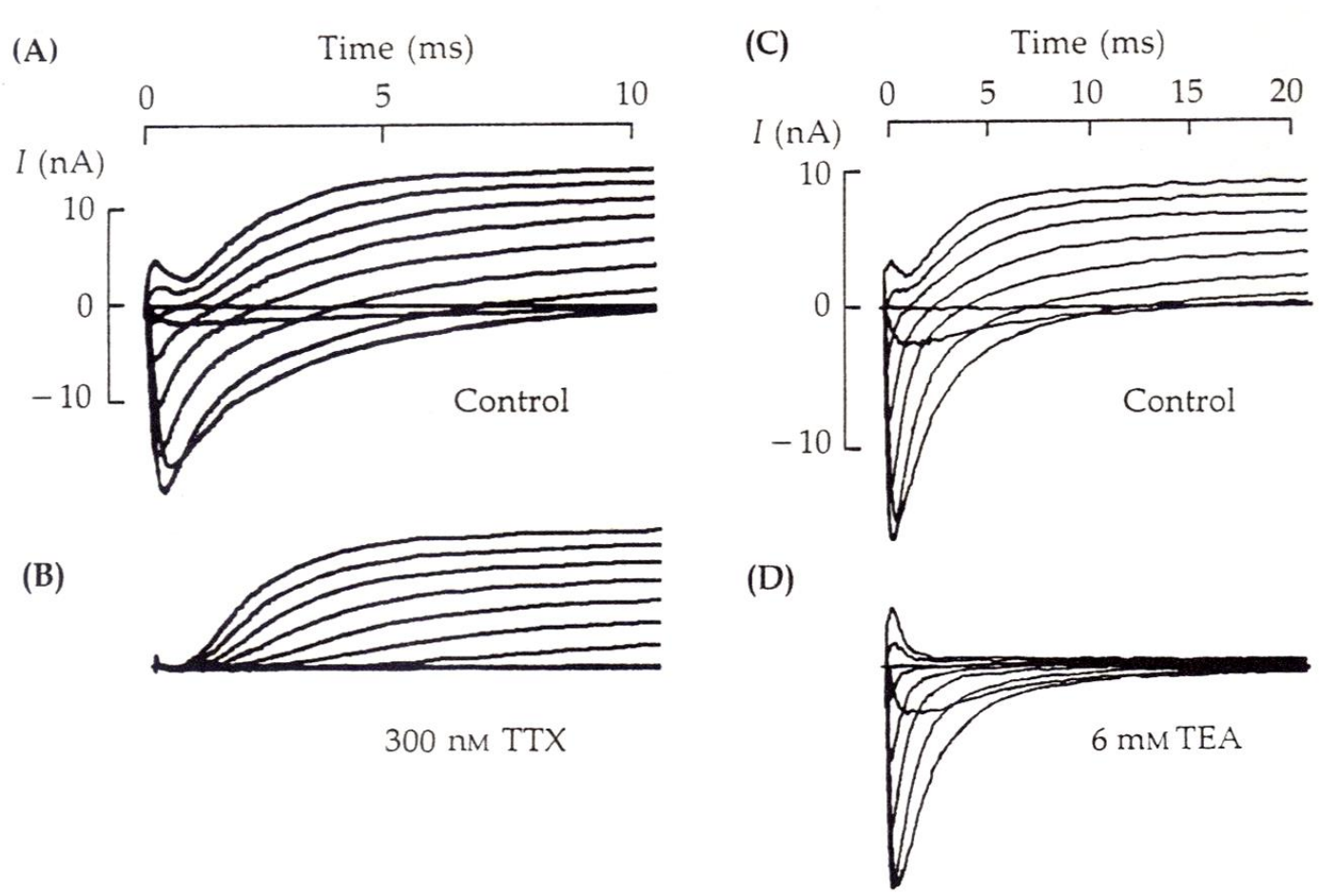


SEPARACIÓN DE CORRIENTES



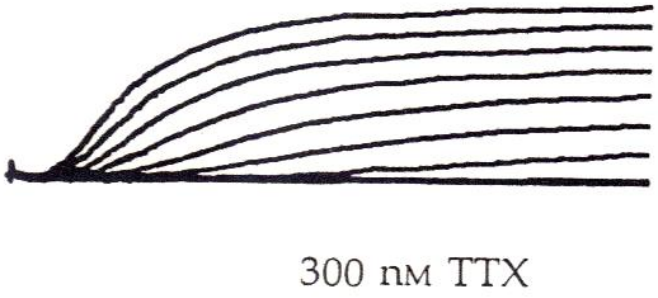


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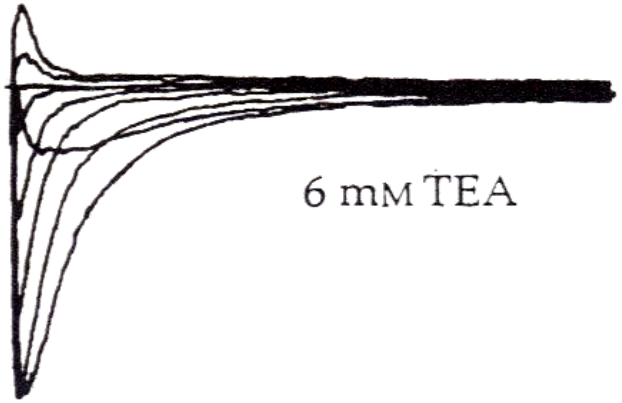


Tomada de “Ion channels of excitable membranes”. B. Hilll. 2ª ed. Fig. 3.2.

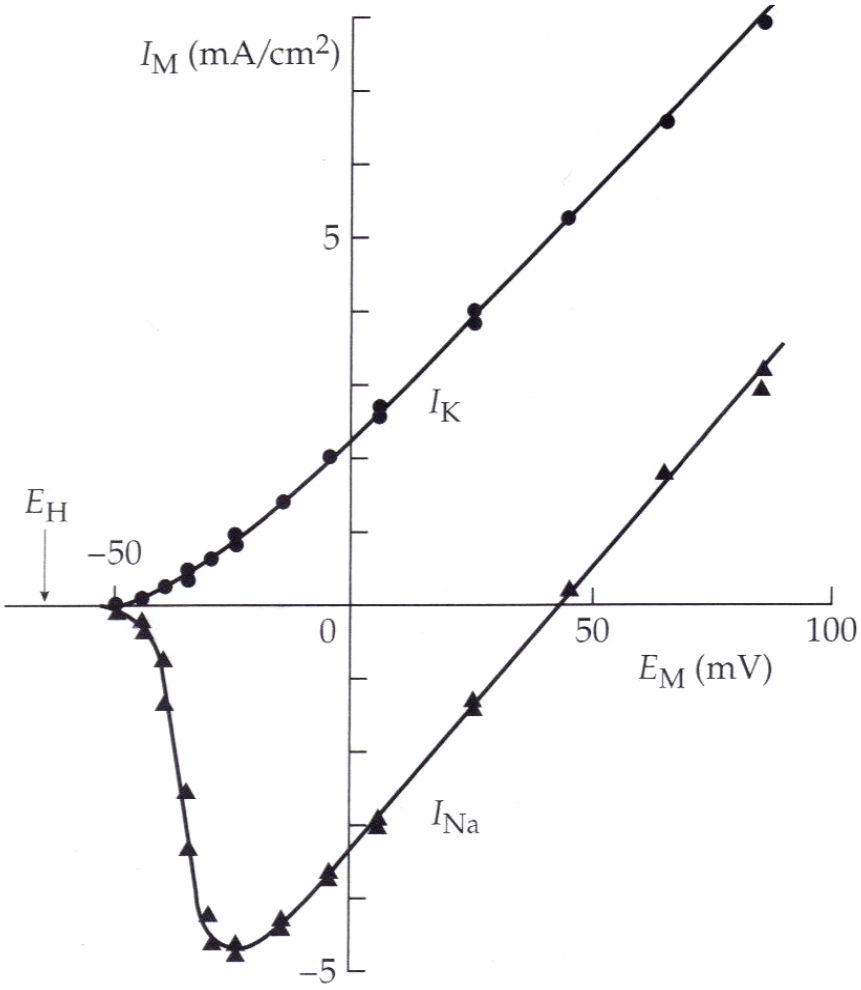
CURVAS I/V DE CORRIENTES



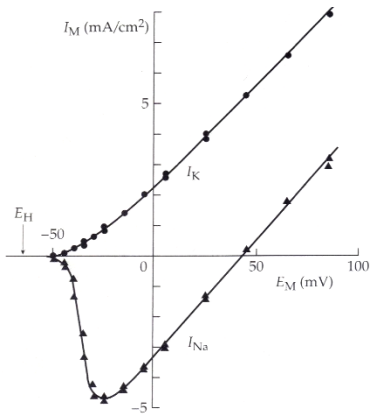
300 nM TTX



6 mM TEA

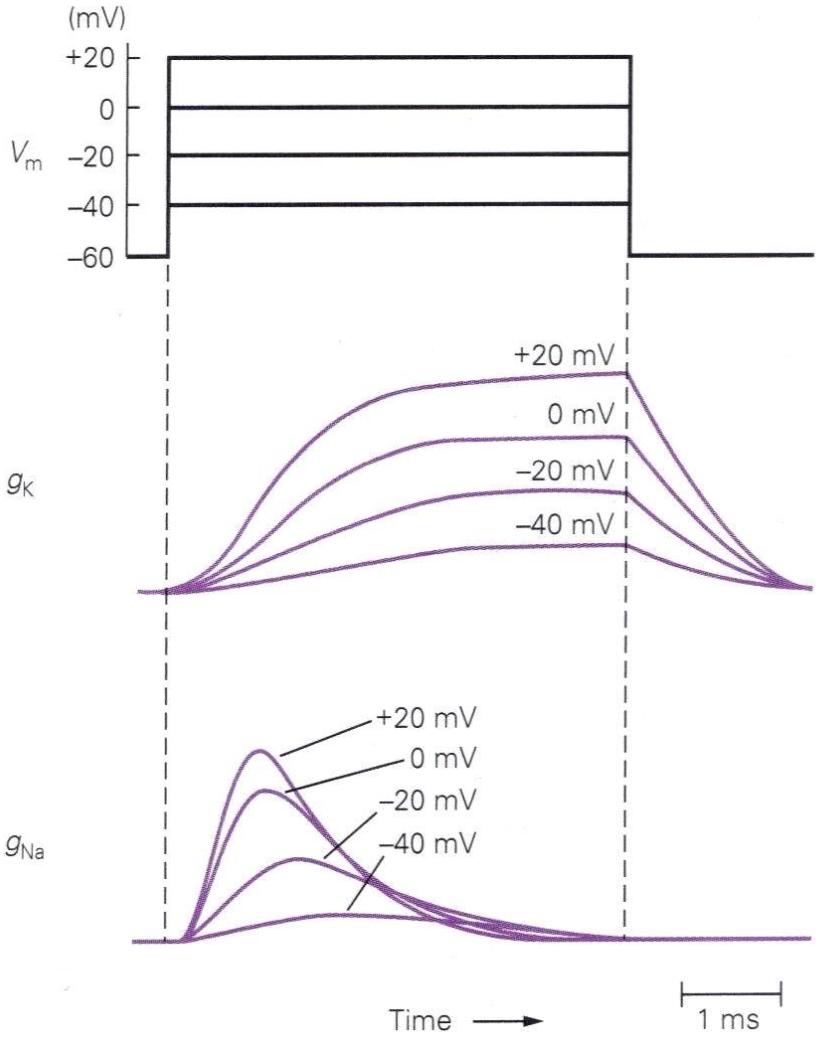


MODIFICACIÓN DE LAS CONDUCTANCIAS

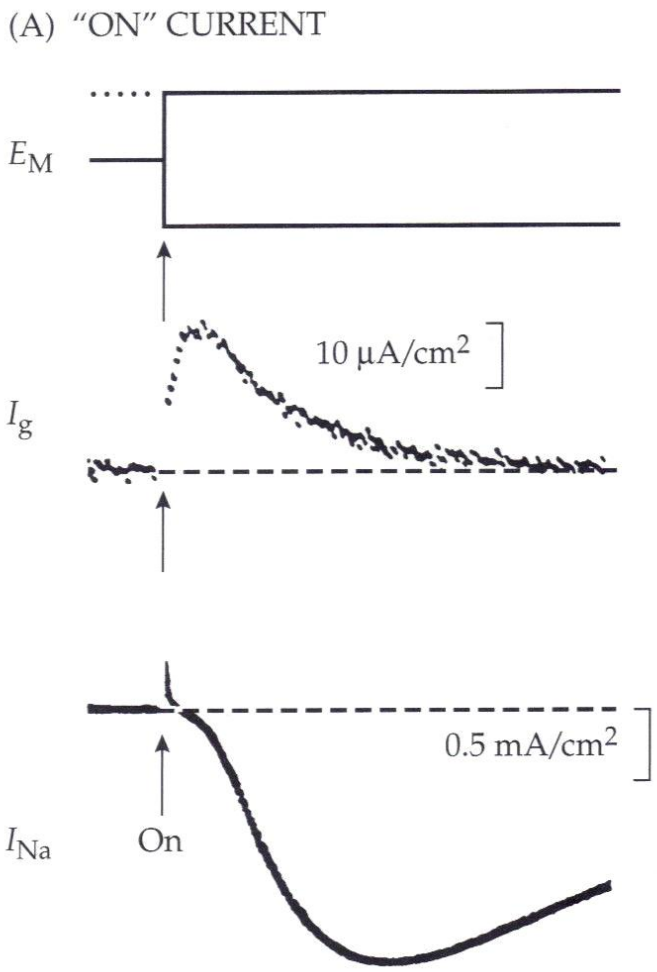


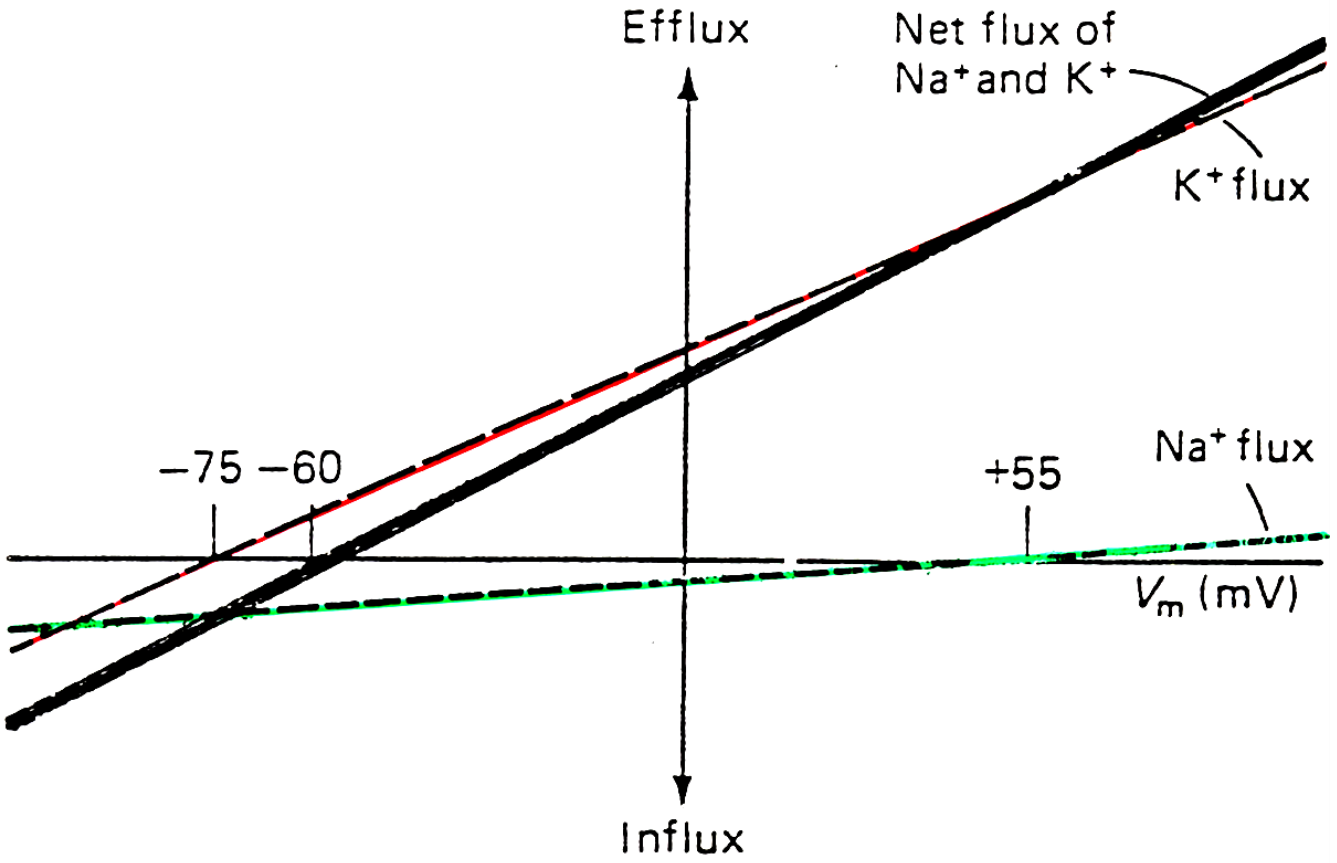
$$g_K = \frac{I_K}{(V_m - E_K)}$$

$$g_{Na} = \frac{I_{Na}}{(V_m - E_{Na})}.$$

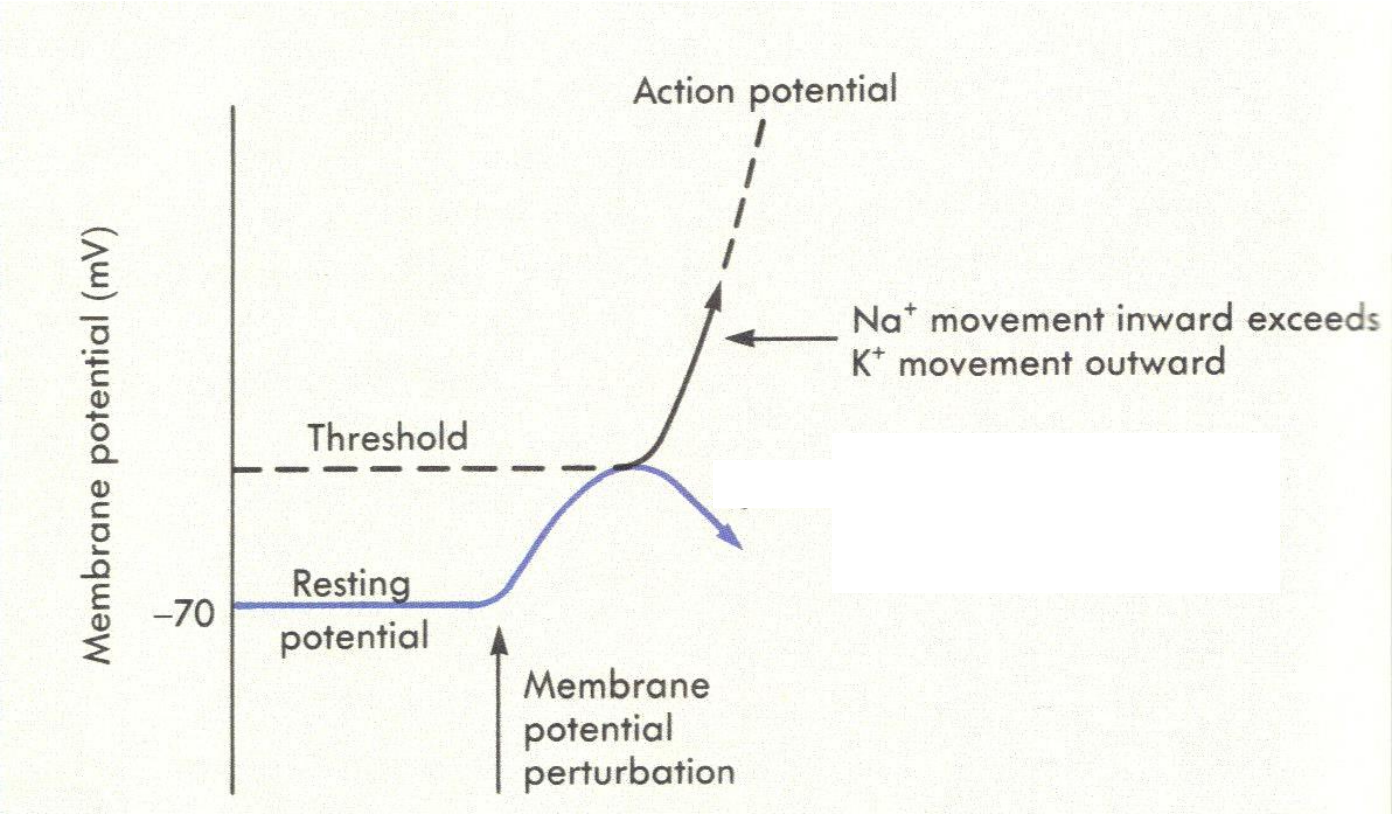


CORRIENTE DE COMPUERTA

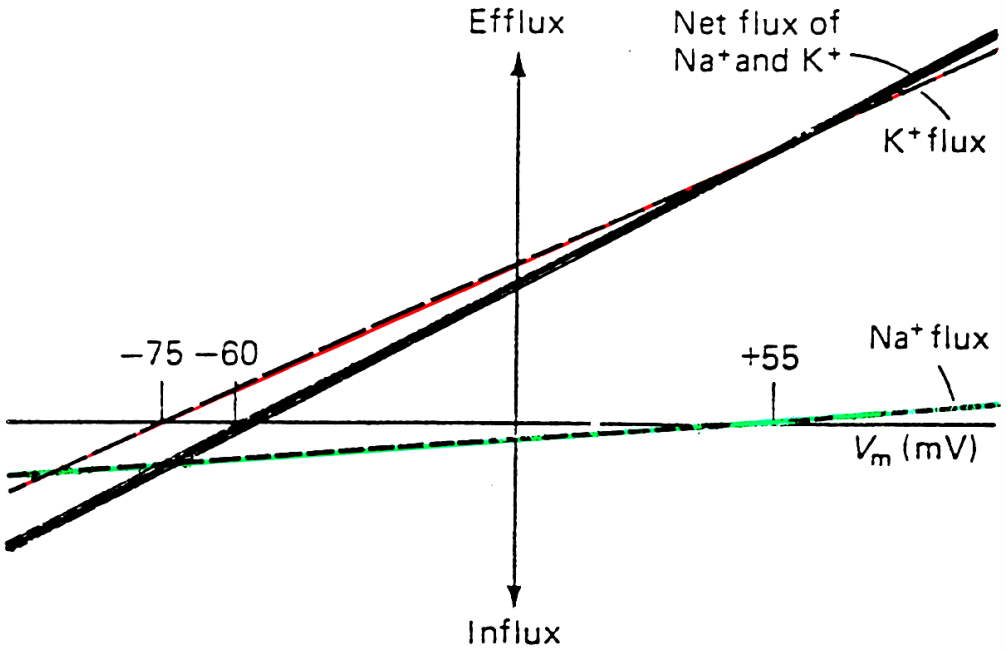


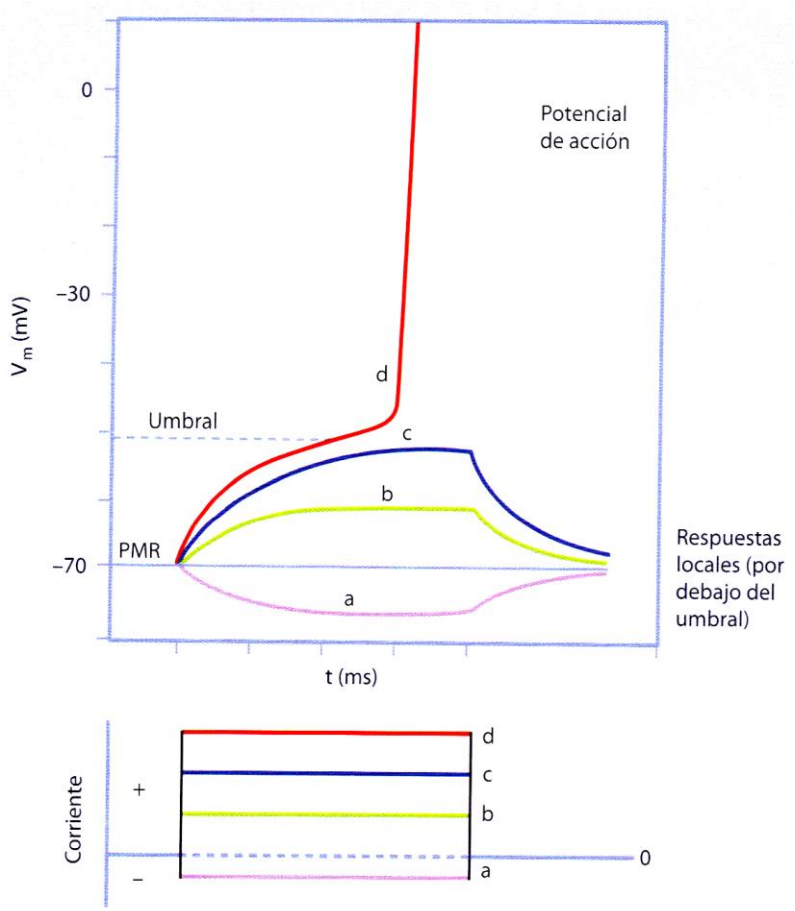

Efflux
Net flux of
Na+ and K+
K+ flux
−75 −60
+55
Na+ flux
Vm (mV)
Influx

UMBRAL DEL POTENCIAL DE ACCIÓN

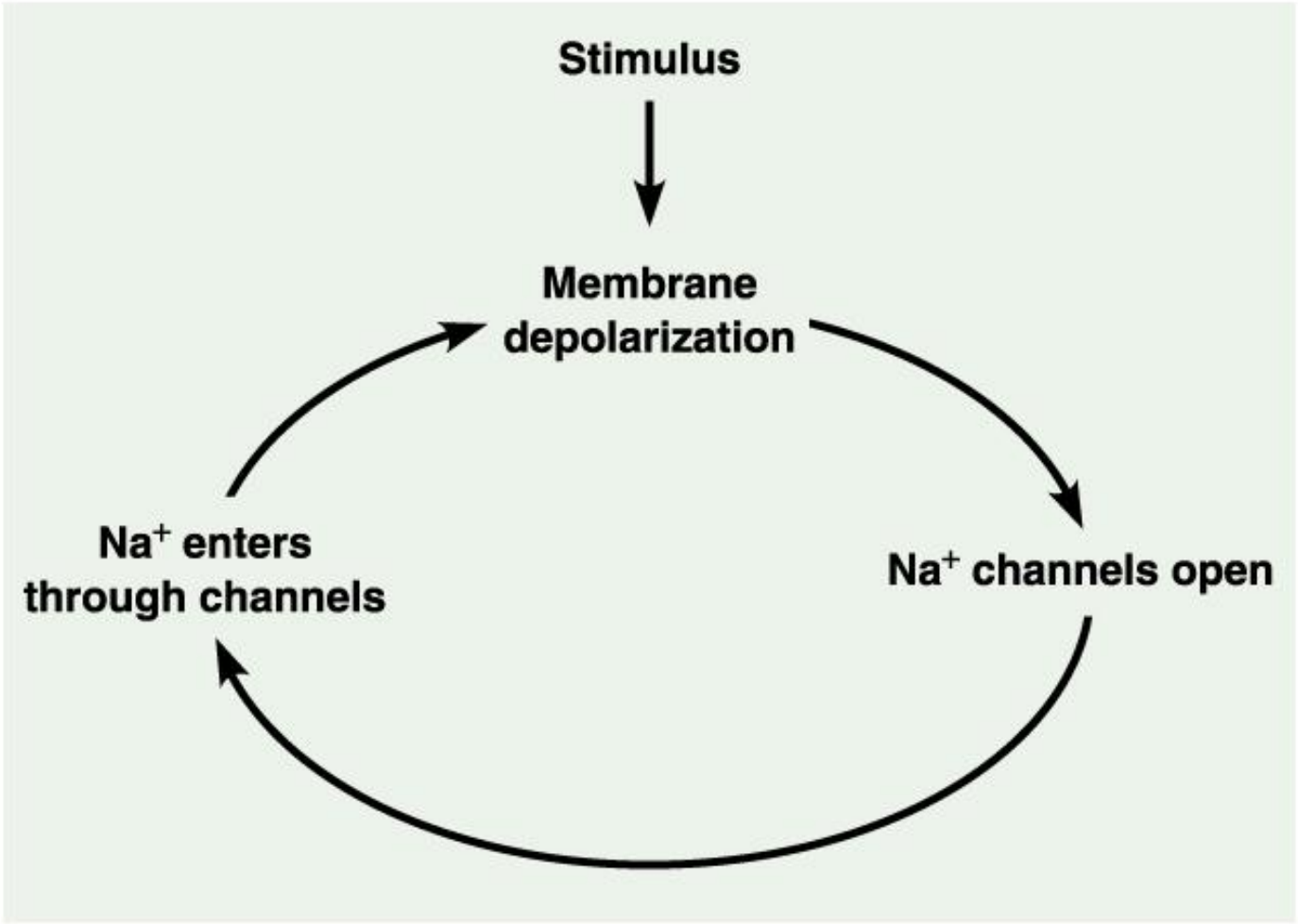


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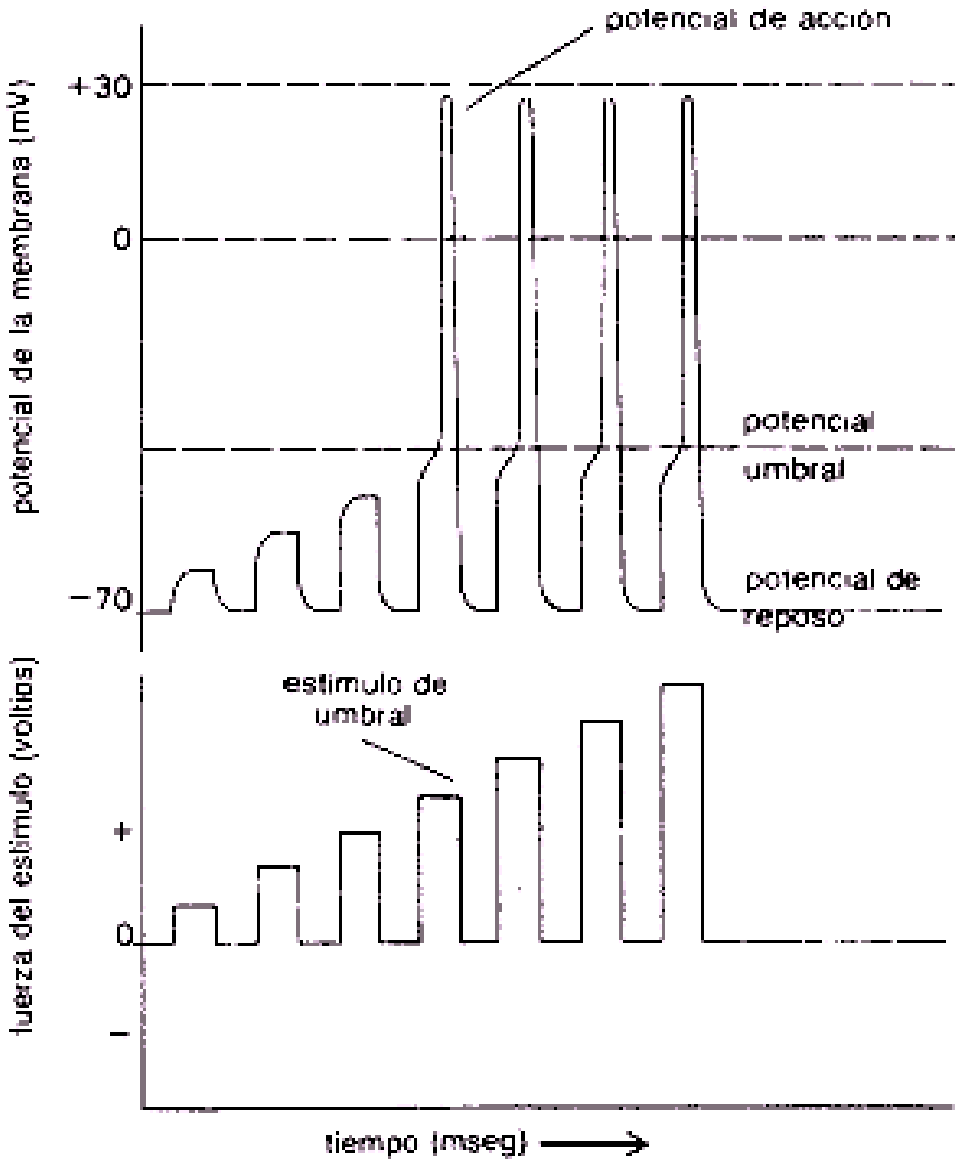
Respuesta autorregenerativa



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Ley del todo o nada

RESPUESTA TODO O NADA



PERÍODO REFRACTARIO

